

5-DOF Robotic Arm

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1 Introduction

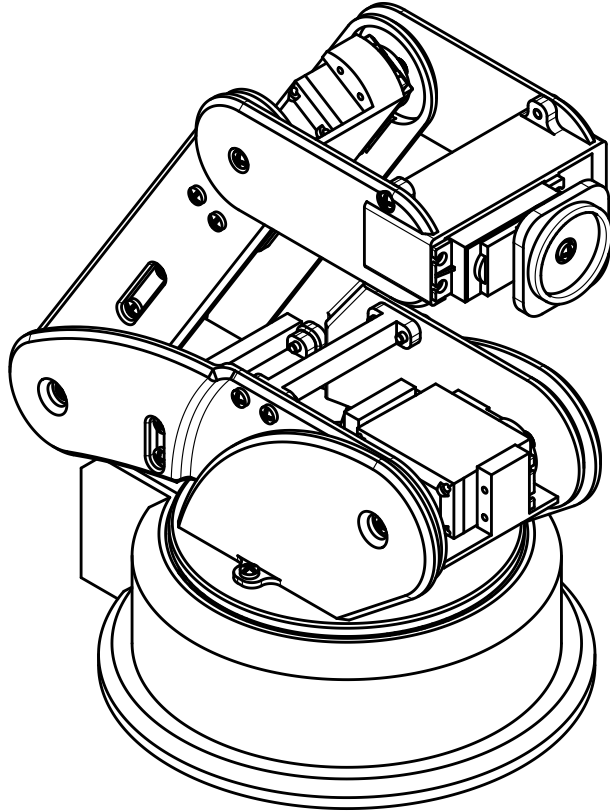


Figure 1: Full arm assembly

This project is a simple yet modular and easily modifiable 5-DOF robotic arm intended to be a platform for the teaching and learning of basic robotics control systems. The project uses five MG996R servo motors as they are a common hobbyist robotics component. Figure 1 shows the final assembled arm in a resting position.

The arm does not possess an actuated gripper as it has been designed with each servo placed directly within the arm's joints in order to avoid the complexities of gearboxes or belt drives. This allows for the stated focus on control systems and means that any issues encountered in actuation of the arm can be immediately attributed to the servos themselves, streamlining the physical troubleshooting process.

2 Drawings

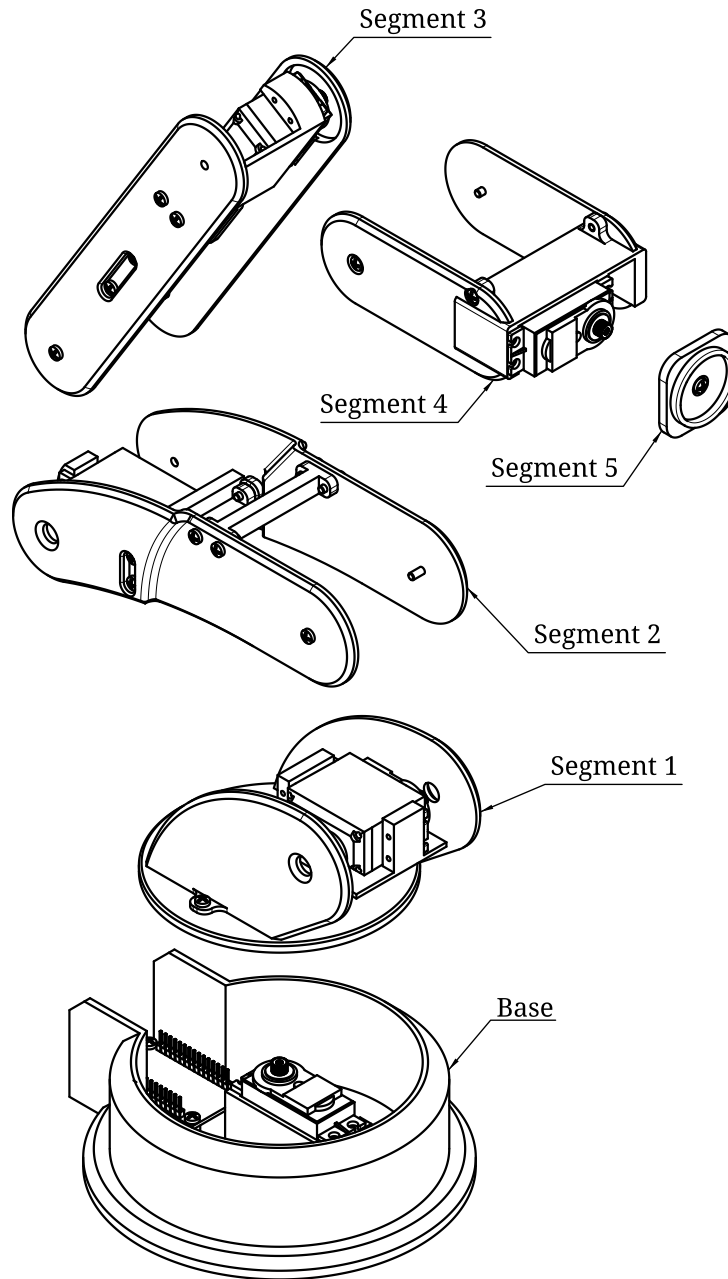


Figure 2: Exploded arm assembly

The arm is comprised of six assemblies. As can be seen in Figure 2, this includes the base of the arm, four link pieces, and a plate at the end of the arm to visualise actuation of the final servo.

The base, shown in Figures 3 is designed to hold the arm's ESP32 microcontroller, power and servo control boards, and any weights that may want to be added in order to hold the arm in place during movement.

Figures 4, 5, 6, and 7 show the construction of each link of the arm, and figure 8 shows the construction of the final "plate" link.

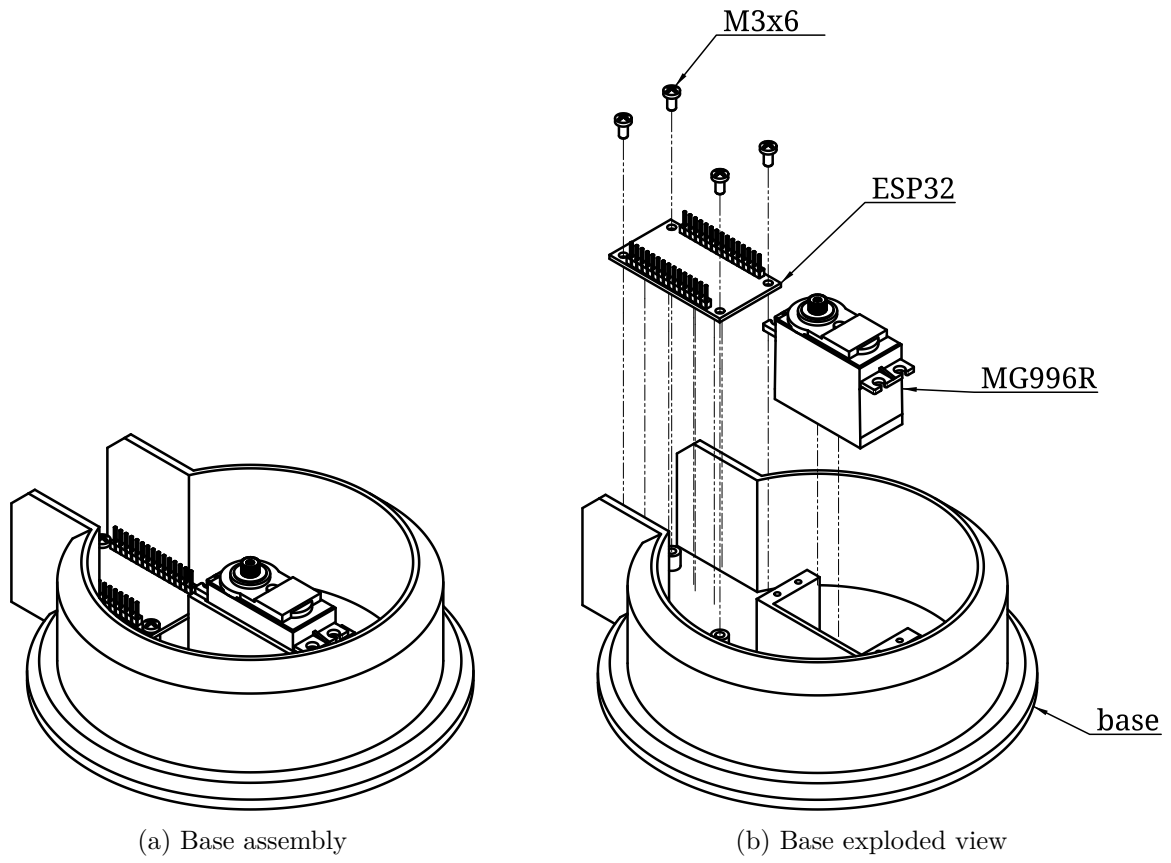


Figure 3: Base views

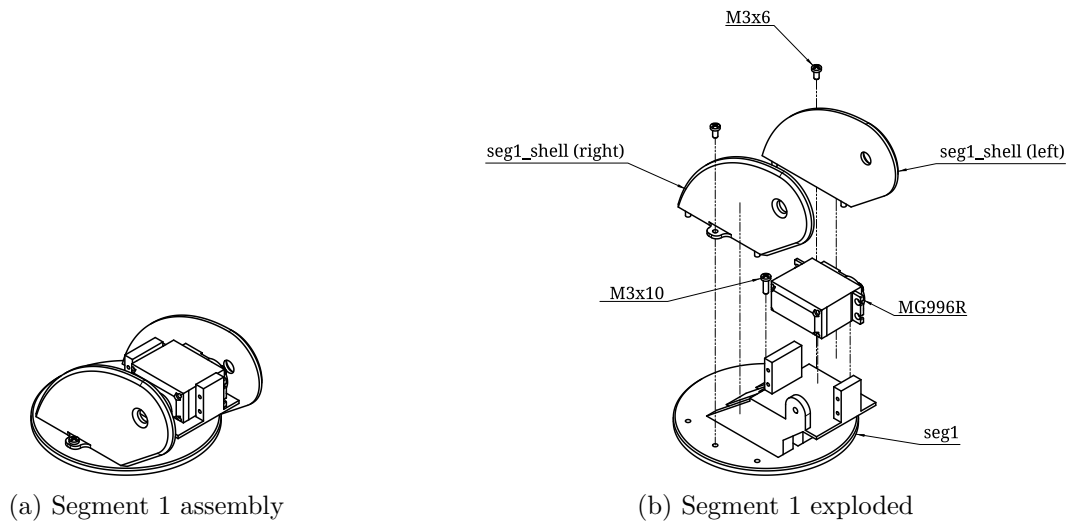
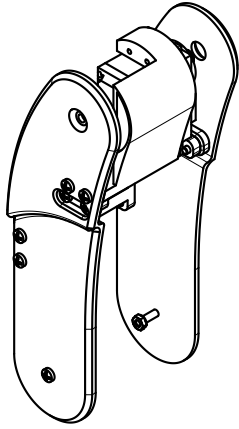
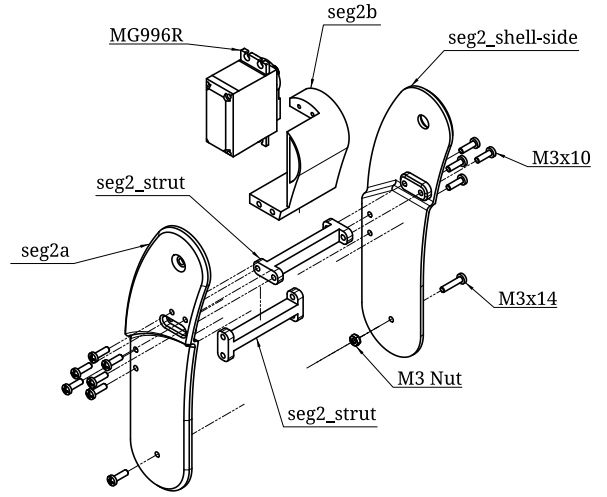


Figure 4: Segment 1 views

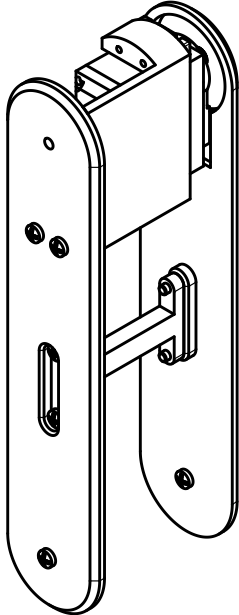


(a) Segment 2 assembly

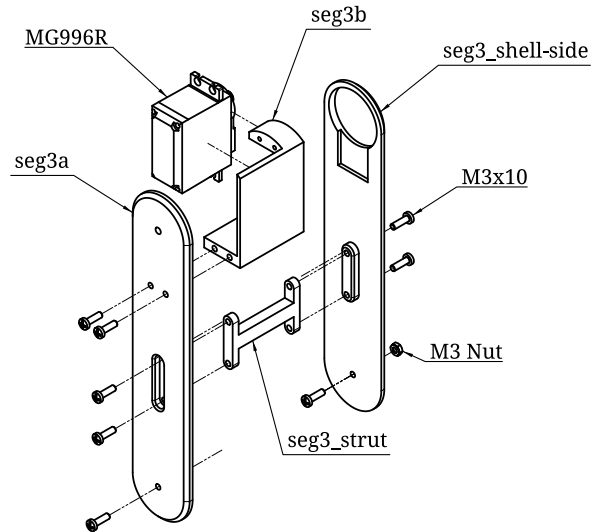


(b) Segment 2 exploded

Figure 5: Segment 2 views

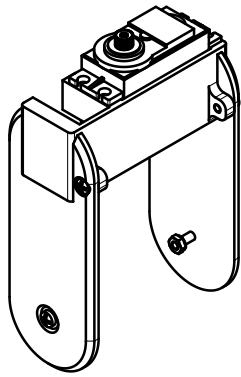


(a) Segment 3 assembly

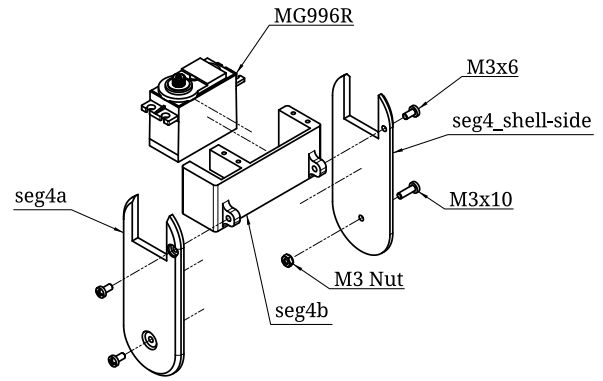


(b) Segment 3 exploded

Figure 6: Segment 3 views



(a) Segment 4 assembly



(b) Segment 4 exploded

Figure 7: Segment 4 views

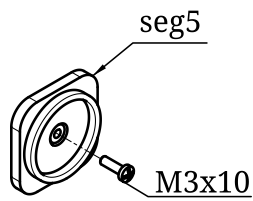


Figure 8: Segment 5 views

3 Bill of Materials (BOM)

Table 1: Purchased Parts

Description	Qty
MG996R Servo	5
ESP32 Devkit Type-C	1
PCA9685 16-Channel 12-Bit PWM Driver	1
DC5525 to Screw Terminal Connector DC Power Jack (Female)	1
10A Adjustable DC-DC Step Down Converter	1

Table 2: Manufactured Parts

Part	Material	Qty
base	PLA - White	1
seg1	PLA - White	1
seg2a	PLA - White	1
seg2b	PLA - White	1
seg3a	PLA - White	1
seg3b	PLA - White	1
seg4a	PLA - White	1
seg4b	PLA - White	1
seg5	PLA - White	1
seg1_shell-side_l	PLA - White	1
seg1_shell-side_r	PLA - White	1
seg2_shell-side_r	PLA - White	1
seg3_shell-side_l	PLA - White	1
seg4_shell-side_r	PLA - White	1
seg2_strut	PLA - White	2
seg3_strut	PLA - White	1

Table 3: Standard Parts

Description	Qty
M3x6 Pan Head Screw	9
M3x10 Pan Head Screw	22
M3x14 Pan Head Screw	1
M3 Nut	3